CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 7: Coordinate Geometry

Topic-wise short notes with solved examples

With coordinates, lengths and positions can be worked out by calculation. This chapter has two tools: the distance formula and the section formula.

1 Coordinates: a quick recap

- A point in the plane is written (x, y). x is the **abscissa** (distance from the y-axis) and y is the **ordinate** (distance from the x-axis).
- A point on the x-axis has the form $(x, 0)$. A point on the y-axis has the form $(0, y)$. The origin is $(0, 0)$.
- Signs in the quadrants: I $(+, +)$, II $(-, +)$, III $(-, -)$, IV $(+, -)$.

2 The distance formula

- The distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- The distance of $P(x, y)$ from the origin is $\sqrt{x^2 + y^2}$.
- The formula comes from the Pythagoras theorem: the horizontal gap is $x_2 - x_1$ and the vertical gap is $y_2 - y_1$.

Example 2.1 Find the distance between $A(3, -4)$ and $B(-1, -1)$.

SOLUTION

$AB = \sqrt{(-1 - 3)^2 + (-1 - (-4))^2} = \sqrt{(-4)^2 + 3^2} = \sqrt{16 + 9}$

$AB = 5$ **units**

Example 2.2 Find the values of k if the distance between $A(3, k)$ and $B(-1, 2)$ is 5 units.

SOLUTION

$\sqrt{(-1 - 3)^2 + (2 - k)^2} = 5$, so $16 + (2 - k)^2 = 25$

$(2 - k)^2 = 9$, so $2 - k = 3$ or $2 - k = -3$

$k = -1$ **or** $k = 5$

Example 2.3 Find the point on the x-axis that is equidistant from $A(3, 4)$ and $B(7, 6)$.

SOLUTION

Let the point be $P(x, 0)$. Then $PA^2 = PB^2$: $(x-3)^2 + 16 = (x-7)^2 + 36$

$x^2 - 6x + 25 = x^2 - 14x + 85$, so $8x = 60$ and $x = 7.5$

The point is $(7.5, 0)$.

REMEMBER

When two distances are equal, equate their **squares**. This removes the square roots at once.

3 Using distances to identify figures

- **Collinear points**: three points A, B, C lie on one line if the sum of two of the distances AB, BC, CA equals the third.
- **Triangle**: isosceles if two sides are equal; equilateral if all three are equal; right-angled if the squares of two sides add up to the square of the third.

Quadrilateral	What to show
Parallelogram	Opposite sides are equal
Rectangle	Opposite sides are equal and the diagonals are equal
Rhombus	All four sides are equal
Square	All four sides are equal and the diagonals are equal

Example 3.1 Show that the points $A(1, 1)$, $B(4, 5)$ and $C(7, 9)$ are collinear.

SOLUTION

$AB = \sqrt{9+16} = 5$, $BC = \sqrt{9+16} = 5$, $AC = \sqrt{36+64} = 10$

$AB + BC = 5 + 5 = 10 = AC$

The points are collinear.

Example 3.2 Show that $A(1, 2)$, $B(4, 6)$ and $C(8, 3)$ are the vertices of an isosceles right-angled triangle.

SOLUTION

$AB^2 = 9 + 16 = 25$, $BC^2 = 16 + 9 = 25$, $AC^2 = 49 + 1 = 50$

$AB = BC$, so the triangle is isosceles.

$AB^2 + BC^2 = 50 = AC^2$, so by the converse of the Pythagoras theorem the angle at B is $90°$.

$\triangle ABC$ is an isosceles right-angled triangle, right-angled at B.

4 The section formula

- The point P that divides the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ **internally** in the ratio $m_1 : m_2$ is

$$P = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

- m_1 goes with the **far** point: m_1 multiplies (x_2, y_2) and m_2 multiplies (x_1, y_1).
- **Mid-point** (ratio $1 : 1$):

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- To **find the ratio** in which a point divides a segment, take the ratio as $k : 1$.

Example 4.1 Find the point that divides the segment joining $A(1, -2)$ and $B(6, 8)$ in the ratio $2 : 3$.

SOLUTION

$$x = \frac{2 \times 6 + 3 \times 1}{2 + 3} = \frac{15}{5} = 3$$

$$y = \frac{2 \times 8 + 3 \times (-2)}{2 + 3} = \frac{10}{5} = 2$$

The point is $(3, 2)$.

Example 4.2 In what ratio does the x-axis divide the segment joining $A(2, -3)$ and $B(5, 6)$? Find the point of division.

SOLUTION

Let the ratio be $k : 1$. On the x-axis the y-coordinate is 0: $\frac{6k - 3}{k + 1} = 0$, so $k = \frac{1}{2}$.

The ratio is $1 : 2$. Then $x = \frac{1 \times 5 + 2 \times 2}{1 + 2} = 3$.

Ratio $1 : 2$; the point is $(3, 0)$.

Example 4.3 Find the points of trisection of the segment joining $A(1, 2)$ and $B(7, 11)$.

SOLUTION

The points of trisection divide AB in the ratios $1 : 2$ and $2 : 1$.

$$P = \left(\frac{7 + 2}{3}, \frac{11 + 4}{3} \right) = (3, 5) \text{ and } Q = \left(\frac{14 + 1}{3}, \frac{22 + 2}{3} \right) = (5, 8)$$

The points are $(3, 5)$ and $(5, 8)$.

Example 4.4 Three vertices of a parallelogram $ABCD$, taken in order, are $A(1, 2)$, $B(4, 3)$ and $C(6, 6)$. Find D.

SOLUTION

The diagonals of a parallelogram bisect each other, so AC and BD have the same mid-point.

Mid-point of $AC = \left(\frac{7}{2}, 4\right)$. Let $D = (x, y)$: $\frac{4+x}{2} = \frac{7}{2}$ and $\frac{3+y}{2} = 4$.

$D = (3, 5)$

5 Common mistakes

- Subtracting in different orders for x and y. Use $x_2 - x_1$ and $y_2 - y_1$, with the same point first each time.
- Sign errors with negative coordinates: $-1 - (-4) = 3$, not -5.
- Pairing m_1 with (x_1, y_1) in the section formula.
- Forgetting to write "units" after a distance.
- Claiming a square after showing only four equal sides. That shows a rhombus; the diagonals must also be equal.

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